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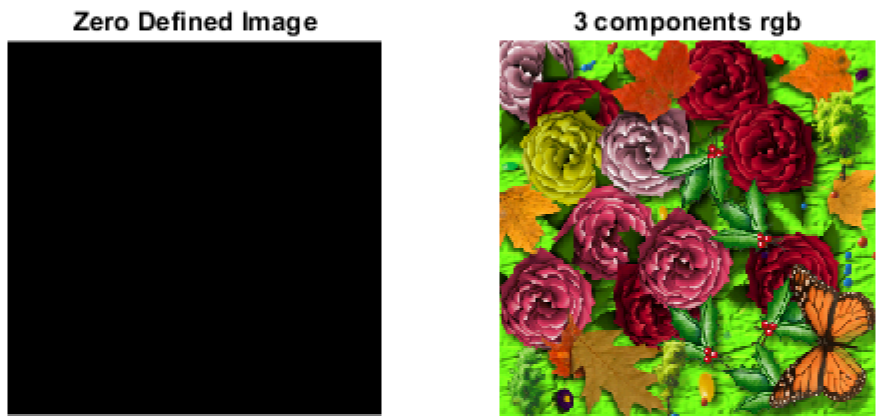
Module 5 Lab 1 - Devesh Parekh and will cox

```
clc
clear
```

Question 1

```
img1=imread('Flower.bmp');
img1_r=img1(:,:,1);
img1_g=img1(:,:,2);
img1_b=img1(:,:,3);
img2=zeros(size(img1));
img2=uint8(img2);
figure(1), subplot(1,2,1),imshow(img2),title('Zero Defined Image');
img2(:,:,3)=img1_b;
subplot(1,2,2),imshow(img2),title('1 component: Blue');
img2(:,:,1)=img1_r;
img2(:,:,2)=img1_g;
subplot(1,2,2),imshow(img2),title('3 components rgb');
```



Question 2

The bright or dark values on each color plane represent the intensity of the color in that pixel.

Question 3

```
img_hsv = rgb2hsv(img1);
% Size of img_hsv
disp(size(img_hsv));
```

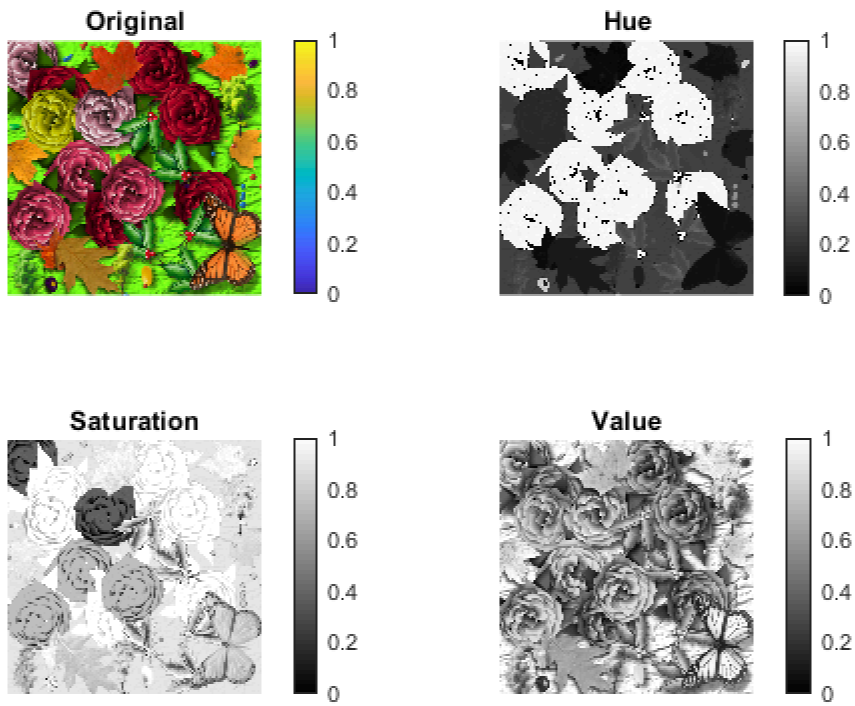
```
   256   256     3
```

Question 4

```
img_hue=img_hsv(:,:,1);
img_sat=img_hsv(:,:,2);
img_val=img_hsv(:,:,3);
figure(2);
subplot(2,2,1)
imshow(img1);
colorbar;

title("Original");
subplot(2,2,2);
```

```
imshow(img_hue);
colorbar;
title("Hue");
subplot(2,2,3);
imshow(img_sat);
colorbar;
title("Saturation");
subplot(2,2,4);
imshow(img_val);
colorbar;
title("Value");
```



Question 5

We have encountered rgb and hsv.

Question 6

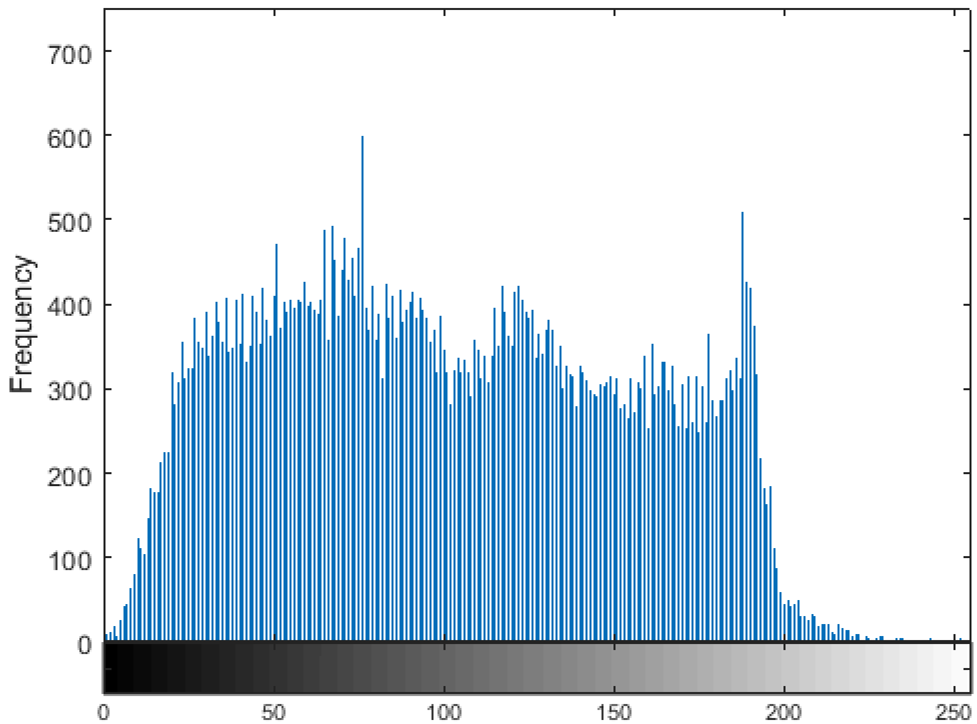
```
img3 = rgb2gray(img1);
% Size of img3

disp(size(img3));
```

```
   256   256
```

Question 7

```
figure(6), imhist(img3);
xlabel("Gray Level");
ylabel("Frequency");
% The histogram shows the frequency occurrence of various gray levels in
% the image.
```

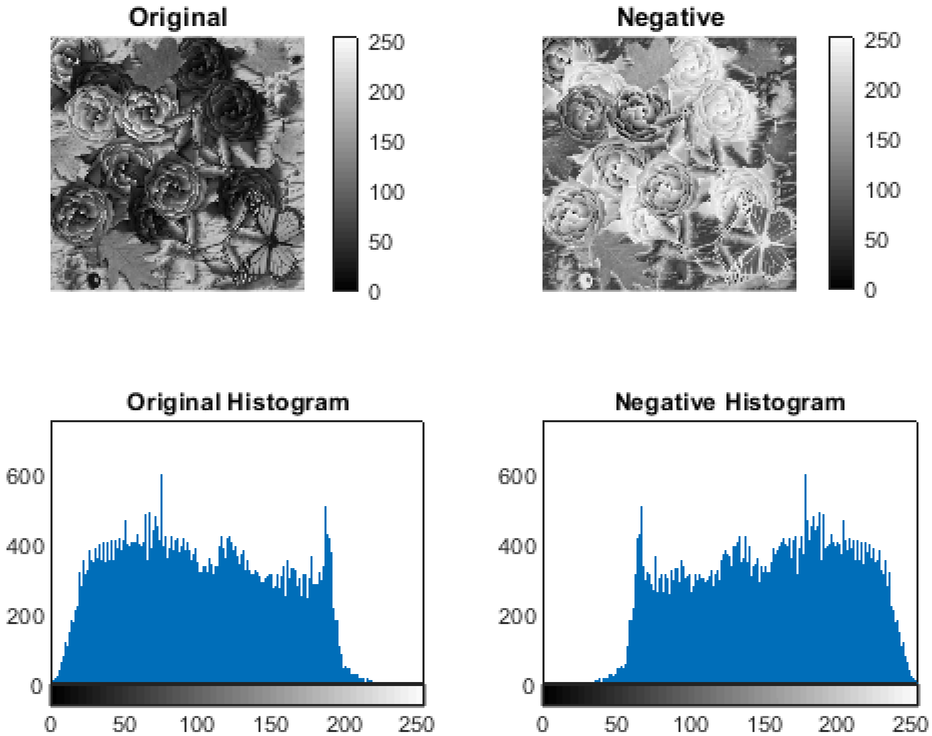


Question 8

```
img3_1=255-img3;
figure(7);
subplot(2,2,1);
imshow(img3);
colorbar;
title("Original");
subplot(2,2,3);
imhist(img3);

title("Original Histogram");
subplot(2,2,2);
imshow(img3_1);
colorbar;
title("Negative");
subplot(2,2,4);
```

```
imhist(img3_1);
title("Negative Histogram");
```

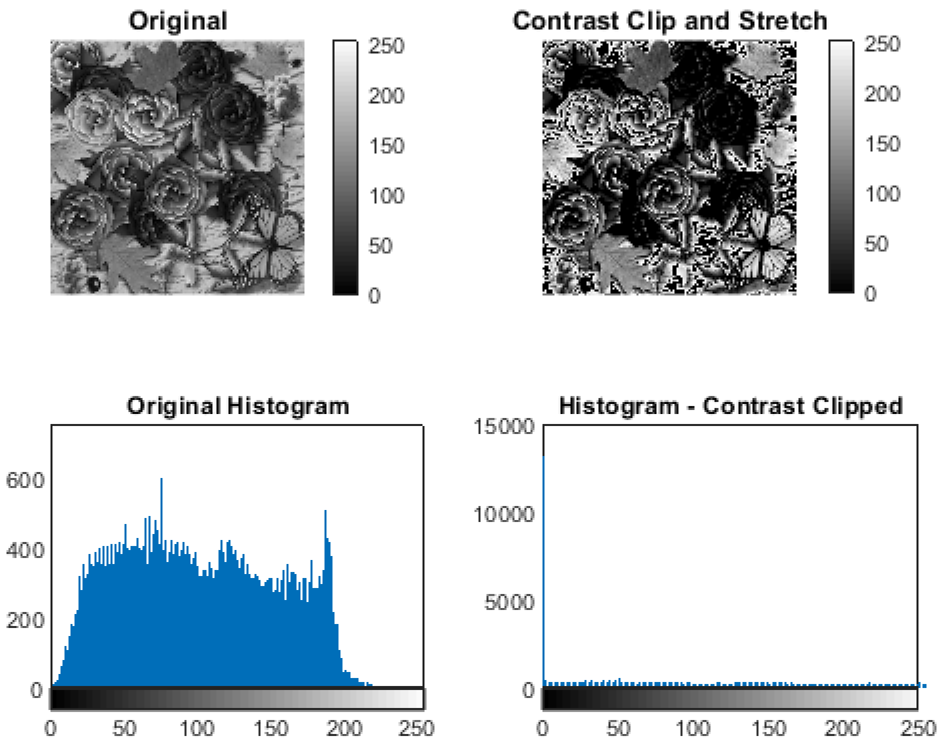


Question 9

```
t1=50;
t2=180;
img3_2=double(img3);
for i=1:size(img3,1)
    for j=1:size(img3,2)
%
        if (img3(i,j)>=t1) && (img3(i,j)<=t2)
            img3_2(i,j)=(img3_2(i,j)-t1)*255/(t2-t1);
%
        elseif (img3(i,j)<t1)

            img3_2(i,j)=0;
%
        elseif (img3(i,j)>t2)
            img3_2(i,j)=1;
        end
    end
end
img3_2=uint8(img3_2);
figure(8);
subplot(2,2,1)
```

```
imshow(img3);
colorbar;
title("Original");
subplot(2,2,2);
imshow(img3_2);
colorbar;
title("Contrast Clip and Stretch");
subplot(2,2,3);
imhist(img3);
title("Original Histogram");
subplot(2,2,4);
imhist(img3_2);
axis([0 250 0 15000]);
title("Histogram - Contrast Clipped");
```



Question 10

There are two spikes at the left and right of the image because we set all values lower than t1 to 0, and all values higher than t2 to 1.

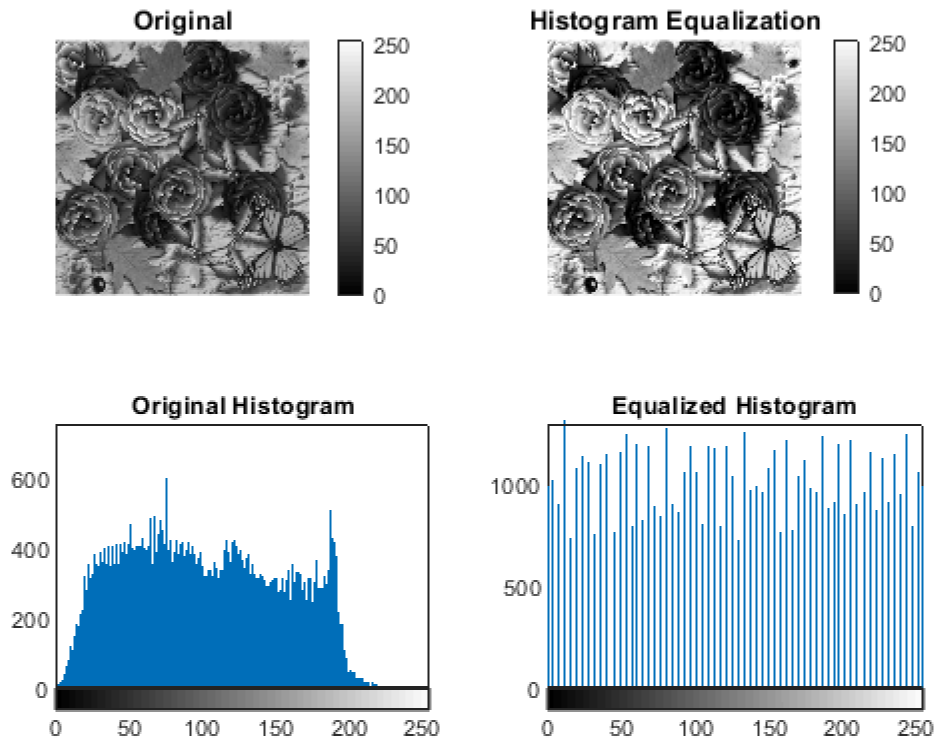
Question 11

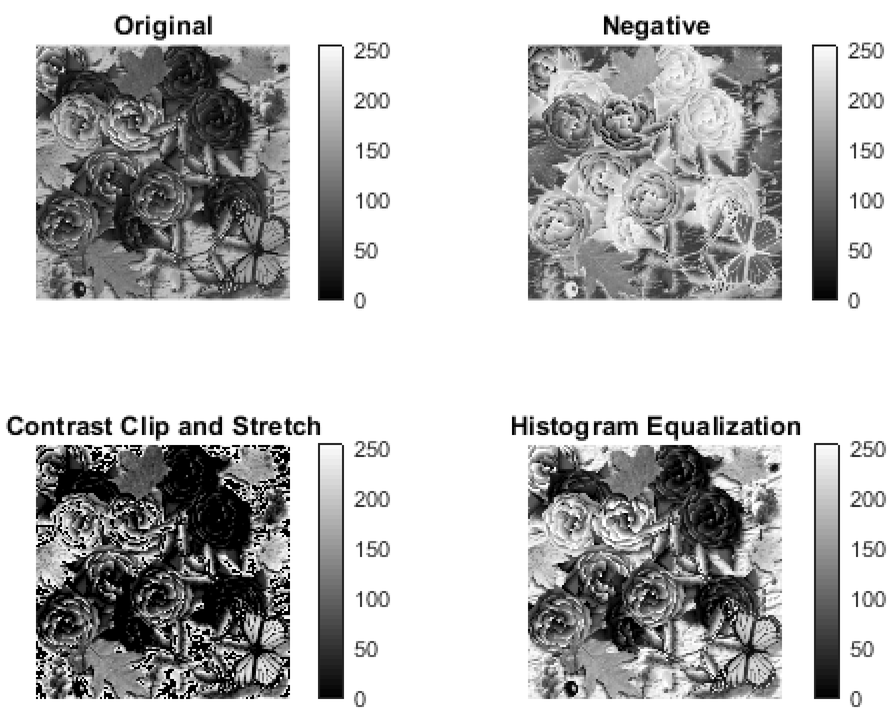
If t1=t2, then yes there would be dividing by 0. You could change the script by adding an if statement that checks if t1=t2.

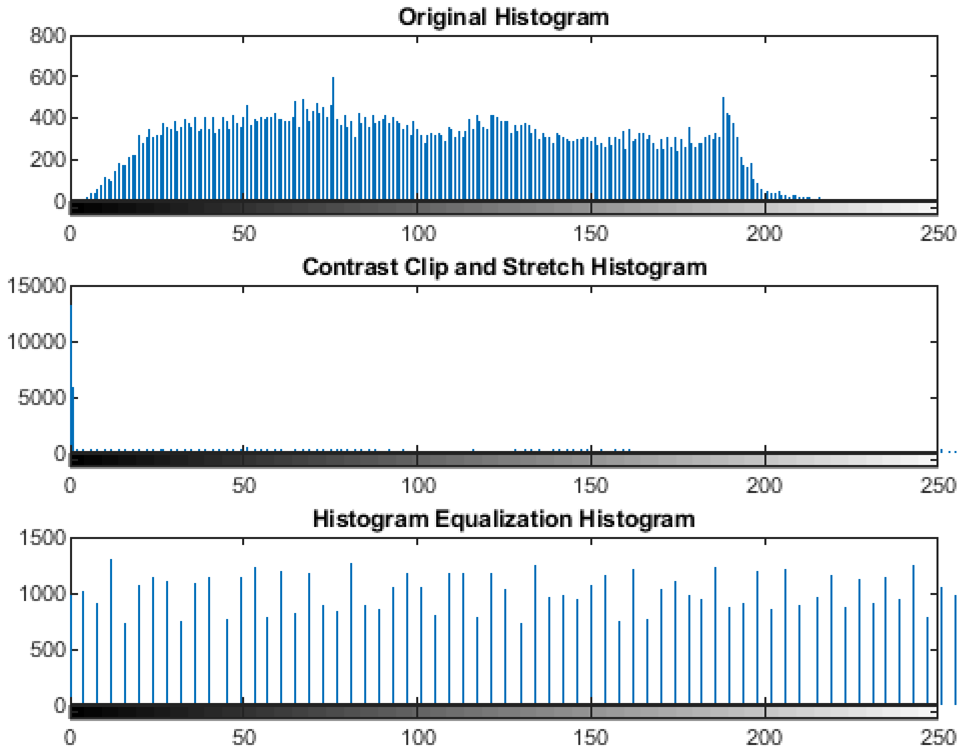
```
img3_4 = histeq(img3);
```

```
figure(9);
subplot(2,2,1)
imshow(img3);
colorbar;
title("Original");
subplot(2,2,2);
imshow(img3_4);
colorbar;
title("Histogram Equalization");
subplot(2,2,3);

imhist(img3);
title("Original Histogram");
subplot(2,2,4);
imhist(img3_4);
title("Equalized Histogram");
% Compare all results
figure(10);
subplot(2,2,1)
imshow(img3);
colorbar;
title("Original");
subplot(2,2,2);
imshow(img3_1);
colorbar;
title("Negative");
subplot(2,2,3);
imshow(img3_2);
colorbar;
title("Contrast Clip and Stretch");
subplot(2,2,4);
imshow(img3_4);
colorbar;
title("Histogram Equalization");
% Compare Histogram results
figure(11);
subplot(3,1,1)
imhist(img3);
title("Original Histogram");
axis([0 250 0 800]);
subplot(3,1,2);
imhist(img3_2);
title("Contrast Clip and Stretch Histogram");
axis([0 250 0 15000]);
subplot(3,1,3);
imhist(img3_4);
title("Histogram Equalization Histogram");
axis([0 250 0 1500]);
```


Original
Histogram Equalization
Original Histogram
Equalized Histogram
0
50
100
150
200
250
200
400
600
500
1000


Original
Negative
Contrast Clip and Stretch
Histogram Equalization
0
50
100
150
200
250

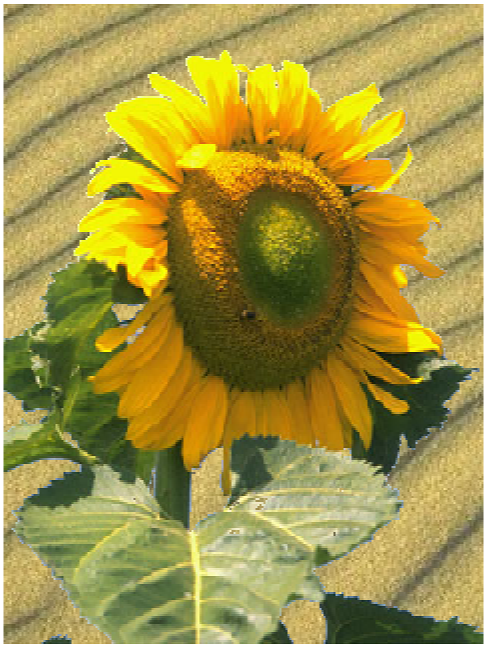


Question 12

Option i

```
sunflower=imread('Images\BlueBG.bmp');
sand=imread('Images\Sand.jpg');
sunflower_r=sunflower(:,:,1);
sunflower_g=sunflower(:,:,2);
sunflower_b=sunflower(:,:,3);
sunflower_final=sunflower;
[row,col]=size(sunflower_b);
for i=1:row
    for j=1:col
        if sunflower_b(i,j)>200
            sunflower_final(i,j,:)=sand(i,j,:);
        end
    end
end
figure(12);
imshow(sunflower_final);
title("Sunflower with Sand Background");
```

Sunflower with Sand Background



Question 13

The background would be put on his shirt since it is recognized as being the same color as the greeclcnscreen. Yes, the special effects system would work, but the weatherman would appear as part of the background.

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